

Global Plastic Pollution & Ocean Health Dashboard

Project Proposal

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GitHub Repository: [Global Plastic Pollution and Ocean Health Dashboard](#)

1 Project Description

This project builds an interactive dashboard that visualizes the global flow of plastic waste — from country-level production and mismanagement, through major polluting rivers, to ocean health degradation. The central question is: **which countries and rivers are the dominant sources of ocean plastic, what structural factors drive mismanagement, and how is ocean health trending as a result?**

The dashboard will guide users through a three-act story: (1) where plastic waste is generated and poorly managed, (2) how it reaches the ocean via river systems, and (3) what impact it has on ocean health across regions.

2 Motivation

Over 8 million tonnes of plastic enter the ocean every year, yet the problem remains abstract to most people. A key insight from research — that just 10 rivers account for roughly 90% of ocean plastic input — gives this dashboard a concrete, actionable story to tell. Plastic pollution sits at the intersection of economic development, infrastructure, and environmental policy, making it highly relevant to Southeast Asia, where several of the worst-offending rivers are located (Mekong, Yangtze, Pasig).

Beyond environmental urgency, this topic is accessible and visually compelling: maps, flow diagrams, and decomposed health scores translate naturally into rich interactive charts.

3 Dataset Description

All datasets are publicly available and free in Table 1. Data spans 2000–2023, is country-level or river-level, and includes numeric, geographic, and temporal dimensions. Known limitations include sparse data for low-income nations and estimation uncertainty in river plastic flow models.

4 Visualization Challenge

This dataset presents several non-trivial visualization problems:

- **Multi-scale spatial mismatch.** Plastic waste is measured at the country level, but ocean impact follows ocean currents and river catchment boundaries — neither of which aligns with political borders. Encoding this requires combining choropleth maps with flow overlays.

Source	Content	Format
Our World in Data — Plastic Pollution	Per-capita waste, mismanagement rates, ocean inputs by country/year	Free CSV
Ocean Health Index	Annual ocean health scores across 10 sub-goals by country	Free CSV
The Ocean Cleanup — River Sources	Top 1,000 plastic-emitting rivers with estimated annual input	Published research data
Kaggle — Plastic Pollution	Country-level waste generation and recycling rates	Free CSV

Table 1: Dataset Sources and Descriptions

- **Causal chain across three levels.** The story runs from land $\rightarrow$ river $\rightarrow$ ocean. Connecting these three spatial scales in a single coherent dashboard without losing the user requires careful layout and linked views.
- **Right-skewed distributions.** A handful of countries and rivers dominate the data. Standard linear scales hide the long tail; log scales confuse general audiences. Interactive filtering is essential.
- **Composite index decomposition.** The Ocean Health Index score is a weighted average of 10 sub-goals. Showing both the overall score and its components — and letting users compare countries — requires part-whole chart design beyond a simple bar chart.
- **Temporal + spatial together.** Showing how mismanagement rates and ocean health evolve over 20+ years, spatially, requires animated or brushable time controls.

5 Wireframe

Layout overview. The dashboard is a single scrolling page with three sequential sections — Act 1 (Land), Act 2 (River), Act 3 (Ocean) — preceded by a sticky title bar with act tabs and a global filter panel (year range slider, country comparison selector, region dropdown). A consistent color encoding runs through each act (amber/coral for land, blue for rivers, teal/green for ocean) so users orient by section at a glance.

Act 1 — Land. A choropleth world map (A1) shows mismanaged-plastic percentage by country, paired with a horizontal bar chart (A2) ranking the top 15 polluters by absolute tonnes. Hovering either updates a tooltip; clicking a country cross-filters every chart on the page. Below them, a stacked area chart (A3) shows waste trajectory by income group from 2000–2023, with a brushable range that controls the global year filter for the entire dashboard and an animation play button.

Act 2 — River. A zoomable map (B1) plots the top 1,000 plastic-emitting rivers, sized and colored by annual ocean input. A KPI card (B2) anchors the 90% / 10-rivers insight, and a Sankey diagram (B3) flows from source country through river to receiving ocean basin. Hovering

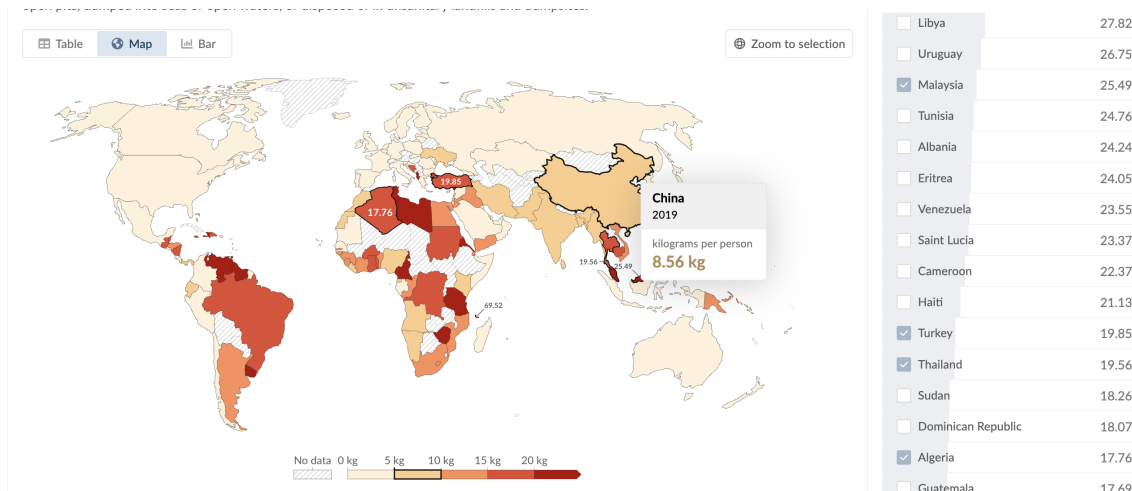


Figure 1: Act 1A

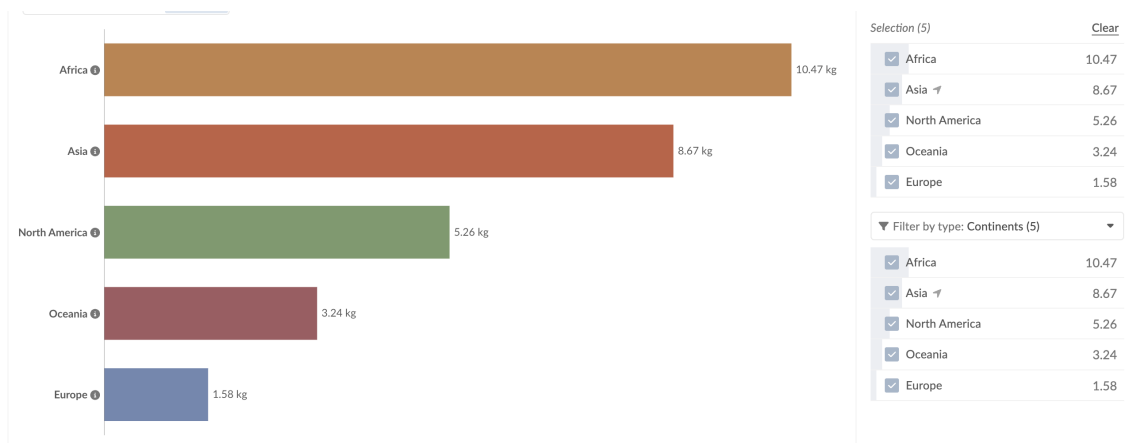


Figure 2: Act 1B

a Sankey path highlights the full chain; clicking an ocean basin scrolls to and filters Act 3.

Act 3 — Ocean. A marine choropleth (C1) renders Ocean Health Index scores by EEZ; a side-by-side toggle swaps it with the Act 1 mismanagement map to overlay cause and effect across the country/marine-region boundary mismatch. A radar chart (C2) decomposes the OHI score into its 10 sub-goals and supports up to three countries overlaid for comparison. A 2×5 grid of small multiples (C3) shows each sub-goal's trend over time, with a shared brushable x-axis and annotation lines for major plastic policy events.

Cross-cutting interactions. A single country selection threads through every act: choosing Indonesia in A1 zooms B1 to Indonesian rivers, filters B3 to flows originating there, and loads Indonesia's profile in C2. The year brush in A3 controls every map's snapshot. Click-throughs on the KPI card and policy markers send context-aware follow-up queries to the dashboard's analysis pane.

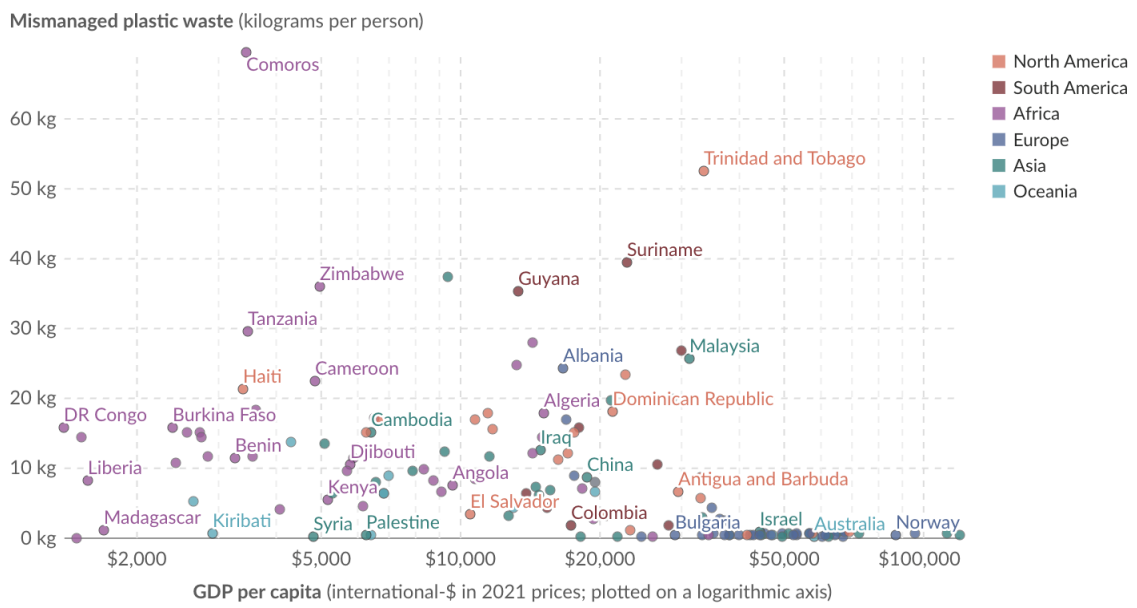


Figure 3: Act 1C

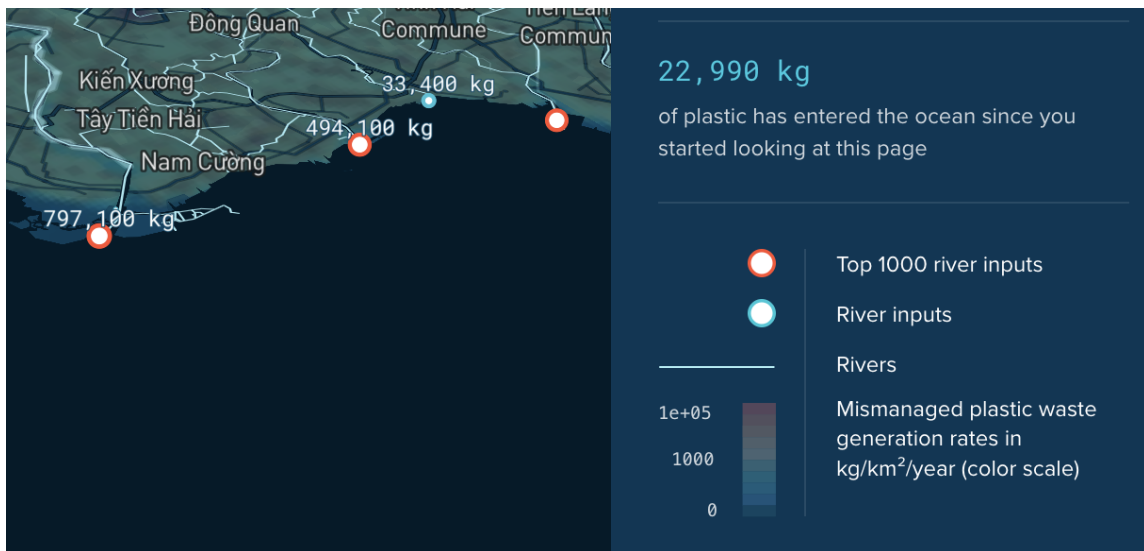


Figure 4: Act 2A

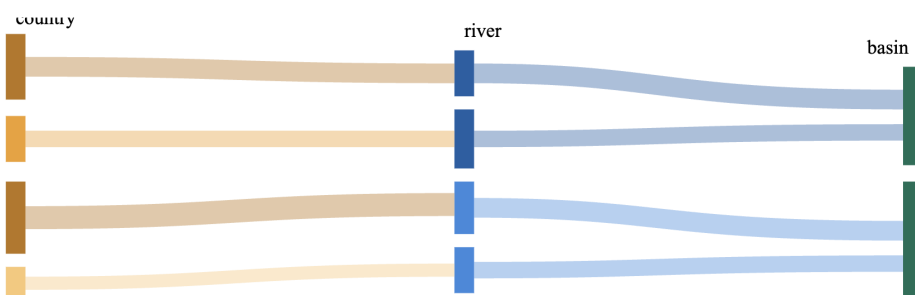


Figure 5: Act 2B



Figure 6: Act 2C

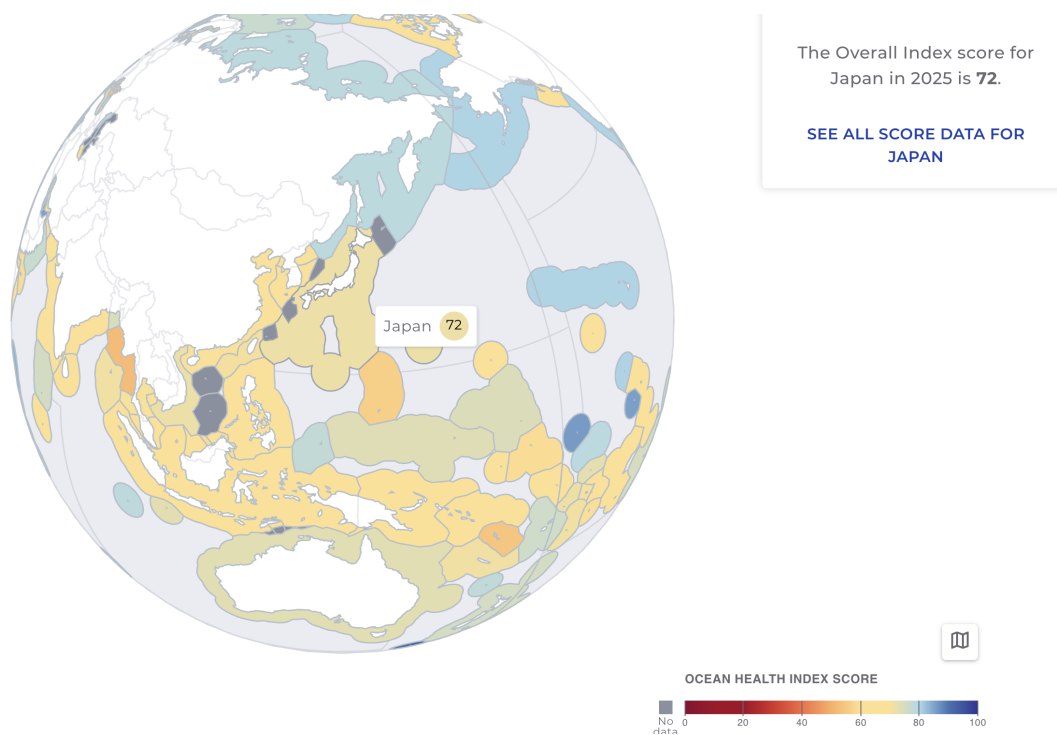
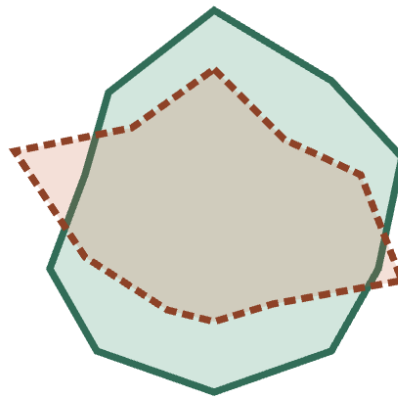


Figure 7: Act 3A



— country A ··· country B

Figure 8: Act 3B

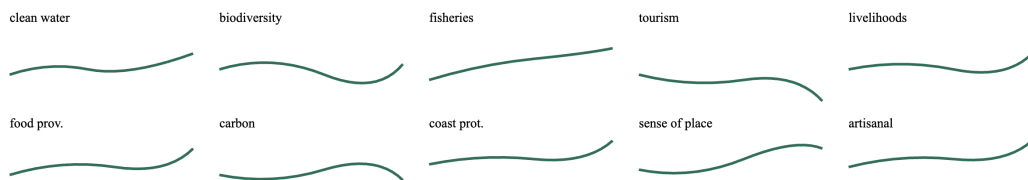


Figure 9: Act 3C